Preconditions

`*this` has an associated asynchronous task.

Effects

As-if `*this=packaged_task(std::move(f))`, where `f` is the stored task associated with `*this`.

Throws

An exception of type `std::bad_alloc` if memory couldn't be allocated for the new asynchronous result.

Std::Packaged_Task::Valid Member Function

Checks whether `*this` has an associated task and asynchronous result.

Declaration

```
bool valid() const noexcept;
```

Returns

`true` if `*this` has an associated task and asynchronous result, `false` otherwise.

Throws

Nothing.

Std::Packaged_Task::Operator() Function Call Operator

Invokes the task associated with a `std::packaged_task` instance, and stores the return value or exception in the associated asynchronous result.

Declaration